

Omar Sadek

Software Engineer — Python, AI/LLM systems, scalable data pipelines

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SUMMARY

Software engineer with deep ownership experience across **Python data pipelines at million-item scale, production AI/LLM systems, and full-stack web at high-growth SaaS scale**. Currently lead engineering on **Instatus Slack Integration** (10M+ visits/month, customers including **Harvard University, Siemens, Sketch, T2, Yum! Brands**), shipping **Wholana** solo at a release-every-two-weeks cadence, and previously founded **Loom Cairo** — a 40,000-MAU marketplace whose backend included a custom **Scrapy crawler ingesting ~1M items / ~4M variants across 43,000 websites in 20 minutes per run**. Strong written communication (English C2; 3x Toastmasters Best Table Topics Speaker).

SKILLS

Languages: **Python** (Scrapy, Flask, TensorFlow, async / concurrency), TypeScript, JavaScript, SQL, C, Bash

AI / LLM systems: Prompt engineering against Anthropic's guidelines, chain-of-thought design, RAG with domain-knowledge documents, **structured-output AI pipelines**, agent skill design, LLM cost / token optimization, Claude, Gemini, TensorFlow

Scalable data pipelines & crawling: **Scrapy at million-item scale** (scheduler, spiders, yielding pipelines, heavy concurrency, proxy rotation), Apify, ETL design, vector embeddings, message queues (BullMQ), durable workflows (Temporal)

Backend: Node.js, Flask, Prisma, ZenStack, multi-tenancy, REST APIs, GraphQL

Frontend: React, Next.js, Tailwind, Shadcn UI

Data: Postgres, Redis, SQLite, vector databases

Cloud & DevOps: AWS (LightSail, S3), DigitalOcean, Cloudflare R2, Linux server admin & SSH, Docker, CI/CD (Vercel, Railway, Buddy, Dokploy), Git, Sentry

Testing: Unit / integration / end-to-end testing, Prisma testing patterns, AI output evaluation

Design Engineering: Framer Motion, Web Animations API, CSS Animations (animations.dev by Emil Kowalski)

EXPERIENCE

Full-Stack Software Engineer — Instatus

Dec 2024 – Present · remote

High-growth status-page SaaS serving **10M+ visits per month** with enterprise customers including **Sketch, T2, Harvard University, Siemens, and Yum! Brands**. Large, complex TypeScript / Node.js codebase.

- **Designed, built, and shipped the Instatus Slack AI bot end-to-end** — a production AI feature now live to all Instatus customers. Prompts engineered against **Anthropic's prompt-engineering guidelines** with explicit **chain-of-thought reasoning**, **RAG grounding against internal domain-knowledge documents**, and aggressive **token / cost optimization** to make the feature economically viable at full GA scale. Evaluated model output quality and edge-case failure modes before shipping.
- **Owned Instatus for Slack end-to-end**, from product strategy and Figma design through implementation: took the integration from a barebones state to a competitive, best-in-class experience, now live to all customers.
- Designed and shipped a **BullMQ message-queue debounce mechanism** for high-frequency backend events, smoothing burst load and reducing redundant downstream work.
- Built durable on-call workflows on **Temporal**, hardening reliability and scheduling guarantees in distributed contexts.
- Designed a **deltas mechanism** for synchronizing large response payloads against a periodically refreshed base log, reducing payload size and bandwidth on incremental updates.
- **Strengthened test coverage on the on-call scheduling system and surfaced multiple previously-undiscovered bugs** through structured edge-case testing.
- Reworked core systems including uptime percentage calculations and incident handling.
- Built features against GraphQL APIs and Redis-backed caching layers; integrated AWS LightSail and S3 for media and asset storage.
- Established Prisma-based unit, integration, and end-to-end testing patterns and contributed to backend reliability and performance optimization.

Founding Engineer — Loom Cairo (later Univyr)

Part of **AUC Venture Lab (V-Lab)** Cycle 25 startup accelerator.

Jun 2023 – Oct 2025

A Python + TypeScript fashion marketplace and SaaS storefront builder for the Egyptian market, built from scratch, serving **40,000+ monthly active users**.

- **Built a high-throughput Python Scrapy crawler ingesting ~1M items / ~4M variants across 43,000 websites in ~20 minutes per run.** Deep work on Scrapy internals — **scheduler, spider, yielding pipelines into the database, heavy concurrency, proxy rotation** — for robust, fault-tolerant crawls at million-item scale. Designed with **SOLID-based scraper classes and custom context managers** for DB lock handling, exception flow, and structured logging.
- Built **ML data labeling and image classification with TensorFlow** to enrich the product catalog; implemented vector-embedding search across heterogeneous catalog data.
- **Built the entire commerce backend from scratch — no Shopify, no off-the-shelf e-commerce OS** — reliably serving orders, addresses, user data, and transactional email for 40,000+ MAU.
- Designed the schema for a full **e-commerce OS** inspired by Medusa.js (not using) and adapted to local constraints: **multi-tenancy** with **ZenStack-based authorization** across two distinct user models (marketplace customers; vendor users tied to organizations inside the storefront-builder OS).
- Provisioned and operated the infrastructure solo: configured **AWS LightSail** for the application, deployed the Scrapy crawler to a **DigitalOcean droplet**, managed both via SSH for deployments, service starts, and ongoing ops.
- Architected the platform: monorepo, Prisma schema, session-based auth.

- Grew to **40,000+ MAU** and **70+ brand partnerships**; accepted into **AUC Venture Lab (V-Lab)** startup accelerator. Tracked MAU, MRR, ARPU, CLV, conversion, retention.

Founder & Engineer — Wholana (pre-launch, ongoing)

Mar 2026 — Present · solo · changelog: wholana.com/changelog

A research and ideation platform for short-form video creators.

- **Built the Wholana Decoder:** a **structured-output AI pipeline** that ingests TikTok videos via Apify scrapers, downloads and parses subtitles, runs Gemini-powered analysis, and emits **structured beat sheets, category taxonomy, and outlier scoring** per video — effectively a decomposition / evaluation pipeline for short-form video content.
- **Shipped 6 substantive product releases solo in 2 months** (independently verifiable on the public changelog).
- Designed and shipped the full surface area solo: Explore (with analytics + server-side filtering), video detail, Lists, Creators directory with per-creator analytics, Ideas Studio, Settings, Billing, plus an admin Control Room and Newsroom.
- Stack: Next.js, TypeScript, Prisma, Supabase (Postgres), Better Auth (OAuth + One-Tap), Polar (subscriptions), Apify, Gemini, Cloudflare R2, Imaginary, Sentry, REST.

Software Engineer (Contract) — UNITAR (UN Agency)

Sep 2024 — Nov 2024

- Built the frontend of a React + TypeScript proof-of-concept web app for managing activities across regions, scales, and stakeholders.
- Led UI design in Figma; collaborated in a cross-functional team with Agile workflows, sprint planning, Git feature branching, and **code review**.

Software Engineering Instructor — ITI

Jul 2023 — Aug 2023

- Taught core programming concepts (loops, conditionals, functions, debugging, program flow) to 30+ high school students.

SA Intern — Dell Technologies · Aug–Sep 2022

- Docker, RedHat Linux, and Data Center Architectures coursework; capstone across AI, DevOps, Testing, and 5G.

Access Transmission Engineer Intern — Orange Egypt · Jul–Aug 2022

- Freed ~15% of bit rate on cell sites by phasing out legacy PDH E1 in favor of VLAN-based networking (NEC, Ericsson NMS).

SELECTED PROJECTS

Argonaut — Freelance Engineer, Designer & Brand Strategist

Engineering & trading firm — HVAC, MEP, fire safety

- Owned product, brand, and engineering end-to-end. The CRM is now in **daily production use by a 10+ person team running 8-hour shifts**.
- Translated MEP domain research into extensible CRM schemas with branching fields that consolidated fragmented spreadsheet workflows into a single robust database.

- Shipped quote/RFQ workflows handling multi-million-dollar pipelines, supplier and consultant tracking, dashboards. Marketing site ranks at the top of search results and surfaces in AI-generated responses.

Argotemp — Freelance Engineer & Designer

Equipment rental & maintenance operations platform

- Built end-to-end rental and maintenance operations system: rental unit availability, maintenance state machines, monthly invoicing, job extensions and renewals.

EDUCATION

Misr International University — 2018–2023

BSc. Electronics & Communications Engineering

Relevant coursework: Microprocessor-Based Systems, Logic Circuits Design, Programming with C++.

Continuing education: CS50x, CS50 SQL, CS50W (Harvard); Motion & Interaction Design (animations.dev by Emil Kowalski); Prompt Engineering for Developers (DeepLearning.AI).

LANGUAGES

- **English** — C2 (fluent / bilingual). **Toastmasters Cairo, 3× Best Table Topics Speaker.**
- **Arabic** — Native.
- **French** — B1 (intermediate).